Second Type

If the equation of a curve C in the xy-Plane is given as $y = f(x)$, $a \le x \le b$ then

$$\int_C P(x,y)\,dx + Q(x,y)\,dy = \int_a^b P[x, f(x)]\,dx + Q[x, f(x)]\,\frac{f'(x)}{dx}$$

If C is given $x = g(y)$ $c \le y \le d$ then

$$\int_C P(x,y)\,dx + Q(x,y)\,dy = \int_c^d P[g(y), y]\,g'(y)\,dy + Q[g(y), y]\,dy$$

If C is given in Parametric form $x = f(t)$, $y = g(t)$
$t_1 \le t \le t_2$

then

$$\int_C P(x,y)\,dx + Q(x,y)\,dy = \int_{t_1}^{t_2} P[f(t), g(t)]\,f'(t) + Q[f(t), g(t)]\,g'(t)\,dt$$

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Ex Evaluate $\int_C (1+x^2y)\,dx + 2xy\,dy$ from $(0,0)$ to $(1,1)$

along the following paths C:

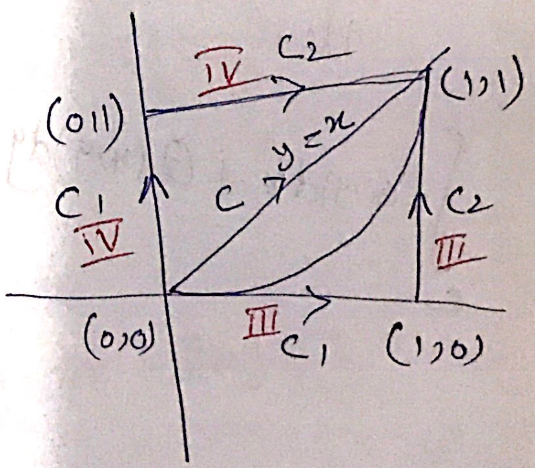
i) the straight line from $(0,0)$ to $(1,1)$

ii) the parabola $y=x^2$ from $(0,0)$ to $(1,1)$

iii) the line segment C_1 from $(0,0)$ to $(1,0)$ and then the line segment C_2 from $(1,0)$ to $(1,1)$

iv) The line segment C_1 from $(0,0)$ to $(0,1)$ and then segment C_2 from $(0,1)$ to $(1,1)$

i) Sol Along the st. line $y=x$ we have $dy=dx$ while x varies from 0 to 1



$$\int_C (1+x^2y)\,dx + 2xy\,dy =$$

$$\int_0^1 (1+x^2\cdot x)\,dx + 2x\cdot x\,dx$$

$$= \int_0^1 (1+x^3+2x^2)\,dx$$

$$= x + \frac{x^4}{4} + \frac{2x^3}{3}\Big|_0^1$$

$$= 1 + \frac{1}{4} + 2/3 = 23/12$$

Eqn of line passing through Two Points $(0,0)$ and $(1,1)$ is $y=x$

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ii Along the Parabola $y=x^2$, we have $\underline{62}$,

$dy = 2x\,dx$ while x varies from 0 to 1

$$\int_C (1+x^2y)\,dx + 2xy\,dy = \int_0^1 (1+x^2\cdot x^2)\,dx + 2x\cdot x^2\cdot 2x\,dx$$

$$= \int_0^1 (1+x^4+4x^4)\,dx$$

$$= \int_0^1 (1+5x^4)\,dx$$

$$= x + \frac{5x^5}{5}\Big|_0^1$$

$$= x + x^5\Big|_0^1 = 1+1 = 2$$

iii In this case

$$\int_C (1+x^2y)\,dx + 2xy\,dy = \int_{C_1} (1+x^2y)\,dx + 2xy\,dy$$

$$+ \int_{C_2} (1+x^2y)\,dx + 2xy\,dy \longrightarrow (1)$$

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Along the line Segment from $(0,0)$ to $(1,0)$,

$y=0$, $dy=0$ while x varies from 0 to 1

$$\int_{C_1} (1+x^2y)\,dx + 2xy\,dy = \int_{x=0}^{1} dx = 1$$

Along the line Segment from $(1,0)$ to $(1,1)$

$x=1$, $dx=0$ while y varies

from 0 to 1

$$\int_{C_2} (1+x^2y)\,dx + 2xy\,dy = \int_{y=0}^{1} 2y\,dy = \frac{2y^2}{2}\Big|_0^1 = 1$$

$$\boxed{\therefore \int_{C} (1+x^2y)\,dx + 2xy\,dy = 1+1 = 2 \quad \text{from eqn (1)}}$$

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iv) Along segment C_1, from $(0,0)$ to $(0,1)$,

$x=0$, $dx=0$ while y varies

from 0 to 1

$$\int_{C_1} (1+x^2y)\,dx + 2xy\,dy = \int_0^1 0\,dx = 0$$

Along segment C_2 from $(0,1)$ to $(1,1)$

$y=1$, $dy=0$ while

x varies from 0 to 1

$$\int_{C_2} ((1+x^2y)\,dx + 2xy\,dy = \int_0^1 (1+x^2)\,dx$$

$$= x+\frac{x^3}{3}\Big|_0^1 = 1+1/3 = 4/3$$

$$\boxed{\int_C (1+x^2y)\,dx + 2xy\,dy = 0+4/3 = 4/3}$$

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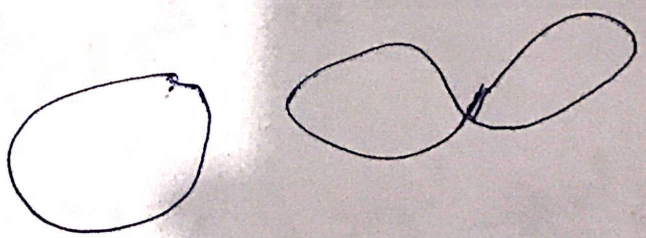
Green's Theorem

Let $P(x,y)$, $Q(x,y)$, $\frac{\partial P}{\partial y}$ and $\frac{\partial Q}{\partial x}$ be continuous functions in a simply-connected region R bounded by a simple closed curve C. Then

$$\oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

where C is described in positive direction. (Counter clockwise) direction.

A Simple closed curve is a closed curve which does not intersect itself any where.



Ex Verify Green's Theorem in the plane for

$$\oint_C (2xy - y^2)\,dx + (x + y^2)\,dy$$ where C is a closed curve of the region bounded by $y = x^2$ and $y^2 = x$

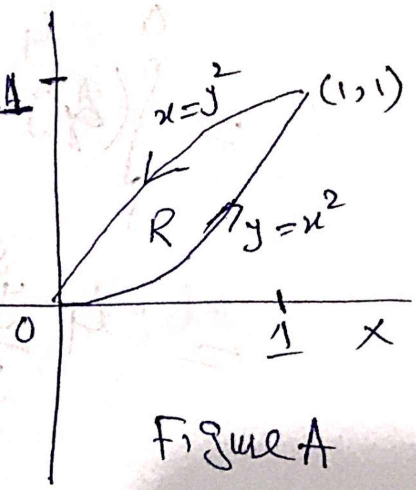


Figure A

Sol
The plane $y = x^2$ and $x = y^2$ intersect at (0,0) and (1,1). The positive direction in traversing C is shown in Figure A.

Along $y = x^2$ we have $dy = 2dx$ while x varies from 0 to 1. Thus the line integral

$$\int_{x=0}^{1} (2x \cdot x^2 - x^2)\,dx + (x + x^4)\,2x\,dx$$

$$= \int_0^1 (2x^3 - x^2 + 2x^2 + 2x^5)\,dx$$

$$= \int_0^1 (2x^5 + 2x^3 + x^2)\,dx = \frac{2x^6}{6} + \frac{2x^4}{4} + \frac{x^3}{3}\Big|_0^1$$

$$= \frac{1}{3} + \frac{1}{2} + \frac{1}{3} = 7/6$$

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Along

$x = y^2$, $dx = 2y\,dy$ while y

Varies from 1 to 0 Thus the line integral

$$\int_{y=1}^{0} (2y^2 \cdot y - y^4)\,2y\,dy + (y^2 + y^2)\,dy$$

$$= \int_{1}^{0} (4y^4 - 2y^5 + 2y^2)\,dy$$

$$= 4\frac{y^5}{5} - \frac{2y^6}{6} + \frac{2y^3}{3}\Big|_1^0$$

$$= 0 - \left(\frac{4}{5} - \frac{1}{3} + \frac{2}{3}\right) = -\frac{17}{15}$$

Thus $\displaystyle\oint_C (2xy - x^2)\,dx + (x + y^2)\,dy$

$$= \frac{7}{6} - \frac{17}{15} = \frac{1}{30}$$

By Green's Theorem in the plane

$$\oint_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

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$P(x,y) = 2xy - y^2$

$Q(x,y) = x + y^2$

$\frac{\partial P}{\partial y} = 2x$

$\frac{\partial Q}{\partial x} = 1$

$$\oint_C (2xy^2 - y^2)\,dx + (x+y^2)\,dy$$

$$= \int_{x=0}^{1} \int_{y=x^2}^{\sqrt{x}} (1-2x)\,dy\,dx$$

$$= \int_{x=0}^{1} y - 2xy \Big|_{x^2}^{\sqrt{x}} dx$$

$\frac{1}{2}+1$

$\frac{3}{2}$

$$= \int_0^1 \left(\left(\sqrt{x} - 2x^{3/2}\right) - \left(x^2 - 2x^3\right)\right) dx$$

$\frac{3}{2}+1$

$\frac{5}{2}$

$$= \int_0^1 \left(x^{1/2} - 2x^{3/2} - x^2 + 2x^3\right) dx$$

$$= \frac{2}{3}x^{3/2} - \frac{4}{5}x^{5/2} - \frac{1}{3}x^3 + \frac{1}{2}x^4 \Big|_0^1$$

$$= \frac{2}{3} - \frac{4}{5} - \frac{1}{3} + \frac{1}{2} = \frac{1}{30}$$

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